SUPERSEDING
MIL-I-81845(AS)
6 January 1972

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED JAU-1/B

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the JAU-1/B cartridge actuated initiator and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-23615	Design and Evaluation of Cartridge Actuated Devices
MIL-C-81805	Cartridge, Impulse, MARK 125 MOD 0

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

FSC 1377

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MIL-STD-129	Marking for Shipment and Storage

DRAWINGS AND ASSOCIATED LISTS

Naval Air Systems Command (CODE IDENT NO. 30003)

DL 275AS100	Initiator, Cartridge Actuated JAU-1/B
DL 2838200	Cartridge, Impulse, MARK 125 MOD 0
DL 298AS116	Closed Chamber Assembly
275AS107	Initiator, Cartridge Actuated JAU-1/B, Packaging Assembly
275AS108	Container, Inner, Assembly

PUBLICATIONS

Bureau of Naval Weapons

WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178	Transportation, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

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3. REQUIREMENTS

3.1 General.

3.1.1 The JAU-1/B cartridge actuated initiator covered by this specification shall be manufactured in accordance with all drawings listed on DL 275AS100.

3.1.2 Materials. All materials used in the manufacture of the initiators shall conform strictly with the specifications referenced on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of initiator manufacture are specified on the drawings, the bidder's selection shall be clearly stated in the proposal.

3.2 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 37 JAU-1/B cartridge actuated initiators shall be manufactured using the same methods and procedures to be used for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purpose of this specification the MARK 125 MOD 0 impulse cartridge is defined as a primary component.

3.3.1 Cartridge. The cartridges shall be furnished from lots manufactured within the preceding 12 months. The cartridges shall be manufactured in accordance with the drawings listed on DL 2838200 and shall be selected from lots which were accepted in accordance with MIL-C-81805. Only primary components from a single lot shall be used in a production lot of initiators; however, one primary component production lot may be used in more than one initiator production lot. See 4.7.1.

3.4 Visual inspection. Initiators shall be free of the following visible defects: Burrs, dents, deep scratches, cracks, illegible markings, corrosion, foreign matter, chipping or flaking of the protective finish, incomplete or damaged threads, and other defects which could adversely affect installation or ballistic performance. See 4.7.2.

3.5 Bridge circuit resistance. The bridge circuit resistance of each completely assembled initiator, when measured electrically as specified in 4.7.3, shall be not less than 1.0 ohm nor more than 2.0 ohms.

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3.6 Cap seating torque verification. A seating torque of 300 ± 10 inch-pounds shall be applied to the caps of the assembled initiators of 3.2, 3.7 and 3.8. The cap shall not move as a result of the applied torque. See 4.7.4.

3.7 Ballistic requirements. Initiators from each production lot, selected in accordance with 4.8.1, shall meet the requirements listed below when subjected to the test prescribed in 4.8.2. An initiator which fails to meet one or more requirements shall be classified as a defective unit.

3.7.1 Pressure. The upper and lower limits of pressure shall be 1250 psi and 650 psi, respectively.

3.7.2 Ignition time. The time between the application of firing current to the initiator and the start of pressure rise in the chamber of the test fixture shall not exceed 0.050 second.

3.7.3 Misfire. The initiator shall fire when proper firing energy is applied. See 4.8.2.

3.7.4 Mechanical failure. All initiator components shall withstand the firings of 4.3.2, 4.8.2 and 4.8.3 with no mechanical failure as defined in 4.8.3.1.

3.8 Locked shut requirements. Twenty initiators from each production lot, selected in accordance with 4.8.1, shall meet the requirements of 3.7.3 and 3.7.4 when subjected to the test prescribed in 4.8.3. An initiator which fails to meet one or more requirements shall be classified as a defective unit.

3.9 Packaging, packing and marking. The packaging and packing of initiators and the container markings shall be in accordance with Section 5. See 4.6.1.

3.10 Workmanship. Initiators shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and assembly operations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 37 JAU-1/B cartridge actuated initiators shall be prepared. Of these initiators, 32 shall be expended in the tests listed below and five shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.1 through 3.6 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-23615: 15 "g" shock (4), vibration (4), and 40-foot drop (6). The numerals in parentheses denote the number of initiators allocated for each test. These initiators (with exception of the 40-foot drop initiators), after subjection to the specified treatments, shall be conditioned for a minimum of six hours at +70° ± 5°F and fired in the test fixture immediately after removal from conditioning. In addition, 18 initiators shall be conditioned for a minimum of six hours, nine at -65° ± 5°F and nine at +160° ± 5°F, and fired immediately after removal from conditioning. Eight of these firings, four from -65°F conditioning and four from +160°F conditioning, shall be conducted in the test fixture. The ten remaining initiators, five from -65°F conditioning and five from +160°F conditioning, shall be fired in the locked-shut condition. All test fixture firings shall be conducted in the test fixture of 4.8.2 and shall meet the requirements of 3.7.1 through 3.7.4. The locked-shut firings shall meet the requirements of 3.7.3 and 3.7.4. The firing current used in all firings shall be as specified in 4.8.2. The failure of any initiator to meet the acceptance requirements as stated above shall cause rejection of the preproduction sample. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk.

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4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 1201 and 10,000 initiators, respectively.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified Minor.

4.6 Quality conformance. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of DL 275AS100 and verification of the following requirements and tests:

Item	Nature of Requirement Verification	Applicable Paragraphs	Classification
Primary component	Single lot	3.3, 3.3.1	(M101)
Cartridge	Age and acceptance	3.3.1, 4.7.1	(M102)
Initiator	Visual inspection	3.4, 4.7.2	(M103)
Initiator	Bridge circuit resistance	3.5, 4.7.3	(M104)
Initiator	Ballistic requirements	3.7, 4.8.1, 4.8.2	(M105)
Initiator	Locked shut requirements	3.8, 4.8.1, 4.8.3	(C1)
Initiator	Packaging, packing and container markings	3.9, 4.6.1	(M106)
Initiator	Cap seating torque verification	3.6, 4.7.4	(M107)

4.6.1 Packaging, packing and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Cartridge verification. Verification that the cartridges used were manufactured and accepted in accordance with 3.3.1 shall be provided.

4.7.2 Visual inspection. Examine all initiators and reject those having any of the defects of 3.4.

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4.7.3 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.5 shall be ascertained by means of a test circuit which limits the bridge current to 25 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate within one percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.5 shall be rejected.

4.7.4 Cap seating torque verification. The test specified in 3.6 is to be performed before the initiators are conditioned and fired. Movement of the scribed index line on the cap and chamber will result in rejection of that initiator. Rejection of one or more initiators will cause rejection of the entire production lot or preproduction sample.

4.8 Ballistic and locked shut tests.

4.8.1 Sampling. A sample of initiators shall be selected at random from each production lot for the ballistic test of 4.8.2 in accordance with the requirements of the single sampling plan of MIL-STD-105, Inspection Level II, and AQL of 0.10. In addition to the sample of initiators for the ballistic test, 30 additional initiators shall be selected at random from each production lot. Of the 30 initiators, 20 shall be fired in the locked shut test of 4.8.3 and 10 shall be retained for investigative purposes.

4.8.2 Ballistic test. The sample of initiators shall be conditioned for a minimum of 6 hrs at +70° ± 5°F and fired in the closed chamber test fixture DL 298AS116. The firing current shall be $5 \begin{smallmatrix} +1.0 \\ -0.0 \end{smallmatrix}$ amperes D.C.

4.8.2.1 Criteria of acceptability. The total number of defective units resulting from the requirements of 3.7.1, 3.7.2 and 3.7.3 shall not exceed the acceptance number specified in Table II-A of MIL-STD-105 for an AQL of 0.10.

4.8.3 Locked shut test. A sample of 20 initiators from each production lot shall be conditioned for a minimum of 6 hours at +70° ± 5°F and shall be fired in a fixture that prevents the gases from venting through the initiator orifice. Firing current shall be $5 \begin{smallmatrix} +1.0 \\ -0.0 \end{smallmatrix}$ amperes D.C.

4.8.3.1 Criteria of acceptability. Any production lot samples (ballistic or locked shut) that exhibit mechanical failure of any metal or plastic initiator component which results in failure to contain the propellant gases shall cause rejection of the lot represented. After firing, the ignition element end of each MARK 125 MOD 0 cartridge of the production lot samples shall be examined for evidence of propellant gas leakage through or around the ignition element. If one or more cartridges of the production lot samples exhibit evidence of propellant gas leakage the production lot shall be rejected.

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4.8.4 Retest. There shall be no retests.

4.8.4.1 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiators, original test results should be discarded and that part of the test reconducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four initiators shall be packaged in an inner container in accordance with Drawing 275AS108.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those initiators which are intended for expenditure in preproduction sample and production lot acceptance tests shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Twelve inner containers as described in 5.1.1 shall be packed in accordance with Drawing 275AS107.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, initiators intended for expenditure in preproduction sample and production lot acceptance tests and packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

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5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on Drawings 275AS108 and 275AS107, respectively. The specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. The JAU-1/B cartridge actuated initiator was originally designed to eject search stores from antisubmarine warfare aircraft.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and, if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. The contract or order should specify that results of ballistic tests are to be forwarded to the Naval Air Systems Command (AIR-53222A) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purpose of this specification, the following definitions apply.

6.4.1 Packaging.

6.4.1.1 Level A. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

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6.4.1.2 Level C. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Packing.

6.4.2.1 Level A. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.2 Level C. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in DOD 4145.26M, "DOD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material".

"Symbols are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes."

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-81845A(AS) Initiator, Cartridge Actuated JAU-1/B

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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